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Research Letter | Psychiatry

Kratom Use Trends Within the Mass General Brigham System From 2017 to 2024

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Introduction

Kratom (*Mitragyna speciosa*) is a plant native to Southeast Asia that exhibits stimulant and analgesic effects, depending on the dosage.[1] Traditionally, it has been used for both medicinal and recreational purposes. Kratom is sold in a variety of forms and can be found in smoke shops and online.

Population estimates of kratom use vary widely. The National Survey on Drug Use and Health[2] reported 1.6 million past-year users in 2023, whereas an upper-bound estimate from Grundmann et al[3] found a 9.1% prevalence in 2024. Kratom-related reports to US Poison Control Centers are also increasing.[4] We sought to identify temporal trends in where kratom was mentioned in care encounters at our health care system using encounter-level electronic health records.

+ Supplemental content

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Methods

For this retrospective cohort study, we used longitudinal clinical data from the Mass General Brigham (MGB) electronic health records to identify emergency department (ED) encounters and hospitalizations that mention kratom from January 2017 through December 2024. The MGB institutional review board reviewed the study and determined the study to be exempt and waived informed consent because data were deidentified. This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline. Kratom mentions in encounters were defined as ED visits or inpatient hospitalizations in which kratom was documented in clinician notes. Identification used a standardized free-text search for the term *kratom*. The unit of analysis was the encounter.

Joinpoint regression version 5.0.2 (National Cancer Institute) was used to assess trends in ED visits and hospitalizations where kratom was mentioned from 2017 to 2024. Models were limited to 0 to 1 joinpoint given limited data points of 8 years. A log-linear model with Monte Carlo permutation testing and bayesian information criterion–based selection was used. Annual percentage change (APC) and average annual percentage change (AAPC) with 95% CIs are reported. The threshold for statistical significance was $P < .05$ in 2-sided tests.

Results

From 2017 to 2024, most ED visits and hospitalizations encounters were with male patients, adults aged 20 to 49 years, and individuals identified as White (**Table**). From 2017 to 2024, cumulative mentions of kratom during hospitalizations increased significantly, with an AAPC increase of 14.9% (95% CI, 8.3% to 22.0%; $P = .001$) (**Figure**, B), whereas kratom mentions in ED encounters increased nonsignificantly (11.3% increase; 95% CI, −4.8% to 33.2%; $P = .16$) (Figure, A). Joinpoint analysis identified nonsignificant increases in kratom mentions ED visit rates from 2017 to 2020 (37.9% increase; 95% CI, −47.8% to 264.2%; $P = .37$) and 2021 to 2024 (7.3% increase; 95% CI, −7.4% to 24.3%; $P = .23$) (Figure, A). Kratom mentions in hospitalization APC increased significantly from 2017 to 2020 (45.4% increase; 95% CI, 17.6% to 93.5%; $P < .001$), followed by a nonsignificant increase from 2020 to 2024 (10.2% increase; 95% CI, −5.1% to 17.2%; $P = .10$) (Figure, B).

Discussion

From 2017 to 2024, the significant increase of encounters with kratom mentioned in our health care system indicates a need for greater clinical awareness. Kratom is commonly used by individuals for self-reported analgesic or mood effects,[3] even though kratom has no US Food and Drug Administration (FDA)–approved medical indications and the US Department of Health and Human Services has concluded that it has no accepted medical use. One concern is the proliferation of products of a kratom alkaloid 7-hydroxymitragynine, a potent μ-opioid full agonist.[5] Kratom remains unscheduled at the federal level, but in July 2025 the FDA recommended 7-hydroxymitragynine be scheduled under the Controlled Substances Act.[5] As kratom and related products become more widely available, the incidence of adverse events may increase, including risk for substance use disorder.[4,6]

Table. Cohort Demographics[a]

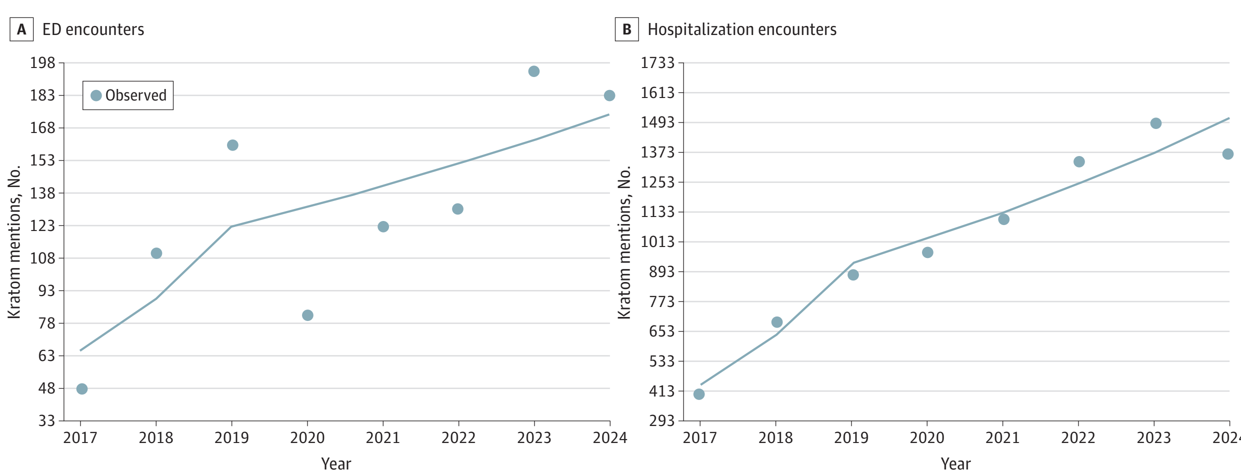
Characteristic	ED encounters, No. (%)	Hospitalizations, No. (%)
Sex		
Female	47 (21.0)	327 (29.8)
Male	177 (79.0)	771 (70.2)
Age group, y		
18-20	3 (1.1)	0
20-29	42 (18.7)	177 (16.1)
30-39	88 (39.3)	309 (28.2)
40-49	20 (9.0)	232 (21.2)
50-59	43 (19.1)	139 (12.7)
60-69	23 (10.1)	155 (14.1)
70-79	6 (2.6)	74 (6.7)
80-89	0	10 (0.9)
Race		
Asian	13 (6.0)	55 (5.0)
Black	16 (7.0)	88 (8.0)
White	152 (68.0)	758 (69.0)
Other[b]	13 (6.0)	55 (5.0)
Unknown	29 (13.0)	143 (13.0)

Abbreviation: ED, emergency department.

[a] Percentages are encounter-level and may use different denominators by variable (sex, age, and race), reflecting institutional privacy masking and rounding. Age percentages are calculated among encounters with age recorded; individuals younger than 18 years were excluded. Documentation of kratom use does not imply causality or attribution of the encounter to kratom.

[b] Included Middle Eastern and multiple races.

Figure. Rates of Kratom Emergency Department (ED) and Hospitalization



Solid line indicates the observed rates of encounters with documented kratom use.

This analysis used free-text extraction of kratom mentions from clinician notes in the electronic health record, which may vary by clinician documentation practices and limits the ability to perform more detailed analyses. Clinicians should routinely inquire about kratom use and inform patients of its risks and benefits. While additional research on kratom's therapeutic potential may be warranted, a focus on the serious adverse effect profile from kratom, 7-hydroxymitragynine, and other kratom alkaloids is urgently needed.

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SUPPLEMENT.

Data Sharing Statement